

SBC 201 Chapter 14 Exterior Walls — Group R-2 High-Rise Dimensional Requirements Matrix (2024)

1. Document metadata and use limitation

- **Project basis:** Riyadh, Saudi Arabia; Group R-2 residential high-rise; an occupied floor is stated to be more than 23 m above the relevant reference level.
- **Deliverable tier:** Project-use matrices in Sections 4–15 (design-check rows, not pasted inventory), plus a coverage summary and unresolved-source register. The full row inventory is not published.
- **Code/source basis:** SBC 201 (2024), Chapter 14, source file Reference\SBC 201 2024\source_reference\Chapter_14 – EXTERIOR WALLS.txt.
- **Extraction audit:** Skill extract. Project-use rows follow the chapter-extract row contract (noun-phrase checks, bold SI values, building-language triggers, named exceptions, check-specific actions). Internal inventory: **225** independently checkable numeric records (**213** Verified, **12** Verify source). Unresolved OCR and the missing ignition table are listed in the register and are not design-release values.
- **Model:** Cursor Grok 4.6.
- **Prepared:** 2026-08-27.
- **Status:** Source-only architectural advisory matrix for design coordination. It is not a stamped compliance statement, facade-engineering report, SCD NOC, SBPS approval, legal opinion, or permission to construct.
- **Outbound-source rule:** No value in this matrix has been imported from Chapter 7, Chapter 16, Chapter 26, Tables 601/602/705.8, SBC 305, SBC 601/SBC 602, IECC climate maps, NFPA 268/275/285 acceptance criteria beyond the numeric triggers stated in this chapter, commentary examples, or the existing chapter summary. Where Chapter 14 sends the user elsewhere, this matrix records the dependency without supplying the outbound value.

Scope and assumptions

1. Group R-2 and high-rise status are project statements, not independently verified classifications.
2. The exact Riyadh AHJ/permit pathway, project stage, fire-strategy status and SCD NOC status are unconfirmed; therefore this matrix does not conclude compliance.
3. Construction type is unconfirmed. An occupied floor above **23 m** is treated as putting combustible WRB, combustible cladding, MCM and HPL onto the Type I–IV **greater than 12 m** path **if** the tower walls are Type I, II, III or IV. Type V framed walls are not assumed for the tower shaft.
4. Automatic sprinkler protection is not selected. Chapter 14 does not branch the cladding numeric limits on NFPA 13 versus 13R.
5. Cladding system, combustible WRB, foam plastic in the envelope, fire separation distance, climate zone and podium mix are unconfirmed.
6. Table 1404.2 stucco coat thicknesses, Table 1404.2 note e, Table 1404.3(3) Zone 8, Table 1404.3(4) and Table 1405.1.1.1.2 are OCR-unresolved or missing from the extract. Affected rows are **Verify source**.
7. Exterior-wall fire-resistance ratings, unprotected opening percentages, foam-plastic NFPA 285 assemblies, veneer design in SBC 305, climate zone from SBC 601/602, and window/door structural tests in Section 1705.9 require separate verification.

2. Legends

Applicability

Status	Meaning
Direct	Expected to govern the stated R-2 tower basis, subject to confirmed geometry and design data.
Conditional	Governs only when the stated feature, cladding system, construction type, climate-zone branch or exception exists.
Not typical	Unrelated occupancy-only or Type V-only rule; omitted from this deliverable unless the gap register already opened that use.
External verification	Chapter 14 points to another section/code/standard, or the project/AHJ basis must be confirmed before use.

Source confidence

Status	Meaning
Verified	Requirement and any stated numeric value were checked against unambiguous mandatory Chapter 14 source text or an unambiguous table cell.
Verify source	OCR, flattened table, page-split, missing table, or footnote attachment is unresolved. Not a design-release value.

3. Project decision and gap register

Decision / gap	Current project basis	Why it controls Chapter 14 application	Required project action
Construction type	Unconfirmed; high-rise R-2 typically Type I or II, not verified	1402.5, 1404.5, 1405.1.1, 1406.10 and 1408.10 apply to Types I–IV; 1406.11 / 1408.12 / 1407.4.1 Type V paths are different	Freeze type of construction on the code data sheet before cladding specification
Cladding system	Unconfirmed: masonry, metal, MCM, HPL, EIFS, wood, vinyl/fiber-cement/PP at podium are all open	Selects which 1404 install rules, 1405 combustible limits, 1406 MCM, 1407 EIFS or 1408 HPL matrix applies	Issue a wall-type schedule by elevation and height above grade plane
Combustible WRB	Unconfirmed	Type I–IV walls greater than 12 m with a combustible WRB require NFPA 285 unless a listed exception is proven	Identify every WRB as combustible or noncombustible per 703.3; do not treat fenestration flashing as the WRB
Foam plastic in the envelope	Unconfirmed	1403.13 and 1406.12 / 1408.13 send foam to Chapter 26; attachment through foam sheathing sends to 2603.11–2603.13	Facade and energy consultants to confirm whether foam is in the wall; do not import 2603 values here
Climate zone	Unconfirmed; IECC / SBC 601/SBC 602 not imported	Tables 1404.3(2)–(4) interior vapor-retarder class and continuous-insulation R-values vary by zone	Lock the SBC 601/602 climate zone for the site; do not assume a zone from maps in this extract
Fire separation distance	Unconfirmed by wall	1405.1.1 10% cap at FSD 1500 mm ; NFPA 268 / Table 1405.1.1.1.2; HPL alternate 1408.11; polypropylene 3 m	Dimension FSD on each elevation; keep Table 1405.1.1.1.2 as a hold point
Combustible cladding height	Occupied floor stated above 23 m; cladding height by material unconfirmed	Types I–IV combustible covering 12 m (FRTW 18 m); MCM/HPL above 12 m need NFPA 285	Mark cladding material versus height above grade plane on every elevation
Podium / Type V framed walls	Unconfirmed	1407.4.1 drainage EIFS is required on Type V framed Group R walls; vinyl/PP/wood siding is a low-height product class	Freeze podium construction type and rainscreen materials separately from the tower shaft

Decision / gap	Current project basis	Why it controls Chapter 14 application	Required project action
Windows and doors	Unconfirmed products	1404.13 sends performance to Section 1705.9; flashing still follows 1404.4	Structural/fenestration schedule to cite 1705.9 tests; wall sections to show 1404.4 flashing
Flood hazard	Unconfirmed	1402.6 / 1402.7 apply only in flood hazard / coastal A / coastal high-hazard areas	Civil/flood consultant to confirm 1612 designation before detailing breakaway walls
NOC and fire strategy	SCD NOC and stamped fire-strategy status unconfirmed	NFPA 285 wall listings, combustible cladding and SCD acceptance cannot be concluded from Chapter 14 alone	Engage the qualified local/fire/facade consultants before design freeze

4. Envelope performance

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1401.1	Chapter scope	This chapter sets minimum requirements for exterior walls, wall coverings, openings, windows and doors, and architectural trim	All exterior walls and coverings	None stated	Direct	Treat Chapter 14 as the weather-envelope and combustible-cladding chapter, not the rating/openings chapter	Verified
1402.2	Weather-resistant envelope	Provide a weather-resistant envelope with flashing per 1404.4, a water-resistive barrier behind the veneer per 1403.2, a means of draining water to the exterior, and condensation protection per 1404.3	Exterior walls	Exception 1: not required over concrete or masonry walls designed to Chapters 19 and 21. Exception 2: ASTM E331 barrier path. Exception 3: EIFS complying with 1407.4.1	Direct	Show WRB, drainage plane and flashing on typical wall sections unless a numbered exception is demonstrated	Verified
1402.2 Ex. 2.1–2.4	ASTM E331 barrier-wall test	Test assembly not less than 1.2 m by 2.4 m , minimum differential pressure 0.3 kN/m² , minimum duration 2 hours , with not fewer than one opening, one control joint, one wall/eave interface and one wall sill; water shall not penetrate those joints	Barrier envelope proposed in lieu of drainage, 1403.2 and 1404.4	Pass/fail is no water at control joints, opening perimeters or dissimilar-material terminations	Conditional	Use only if a tested barrier wall is specified; match tested opening and joint details on shop drawings	Verified
1403.2	Water-resistive barrier	Not fewer than one layer of WRB attached to studs or sheathing, flashed per 1404.4, continuous behind the exterior wall veneer. Comply with No. 15 felt ASTM D226 Type 1; ASTM E2556 Type I or II; ASTM E331 per 1402.2; or other approved materials	WRB required by 1402.2	Concrete/masonry and tested-barrier exceptions under 1402.2	Direct	Specify the WRB product standard on the wall-type schedule and lap it with 1404.4 flashing	Verified
1402.2 Ex. 3	EIFS envelope branch	EIFS complying with Section 1407.4.1 may be used in lieu of the 1402.2 drainage/WRB/flashing prescription	EIFS specified as the exterior envelope	Numeric drainage efficiency is in 1407.4.1, not here	Conditional	Do not treat a barrier-coat EIFS as satisfying 1402.2 unless 1407.4.1 is also met	Verified

5. Combustible water-resistive barriers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1402.5	Combustible WRB NFPA 285	Type I, II, III or IV exterior walls greater than 12 m above grade plane that contain a combustible WRB shall be tested to and comply with NFPA 285. Combustibility per 703.3. Fenestration products, fenestration flashing, WRB flashing and through-wall flashings are not part of the WRB for this section	Type I–IV walls taller than 12 m with a combustible WRB	Exception 1: WRB is the only combustible component and covering is brick, concrete, stone, terra cotta, stucco or steel at Table 1404.2 minimum thickness. Exception 2: WRB is the only combustible component and meets 1402.5 Exception 2.1 and 2.2	Direct	Require an NFPA 285 listed wall assembly for any combustible WRB on the tower shaft, or document a numbered exception	Verified
1402.5 Ex. 2.1	Low-heat WRB cone calorimeter	Peak heat release rate less than 150 kW/m² , total heat release less than 20 MJ/m² , effective heat of combustion less than 18 MJ/kg , ASTM E1354 horizontal, incident flux 50 kW/m² , thickness intended for use	Exception 2 path; WRB is the only combustible component	Must be used together with Exception 2.2	Conditional	Accept this exception only with a complete E1354 report at the installed thickness	Verified
1402.5 Ex. 2.2	Low-heat WRB surface burning	Flame spread index 25 or less and smoke-developed index 450 or less, ASTM E84 or UL 723, specimen mounting ASTM E2404	Exception 2 path; WRB is the only combustible component	Must be used together with Exception 2.1	Conditional	Do not claim Exception 2 from E84/UL 723 alone	Verified

6. Weather covering thicknesses

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.2	Minimum covering thickness	Exterior walls shall provide weather protection. Materials at the minimum nominal thickness in Table 1404.2 are acceptable as approved weather coverings	Exterior weather coverings	Note a: wood siding thinner than 12.5 mm only over sheathing conforming to 2304.6. Note b: thickness is the narrowest solid thickness exclusive of texture. Note d: copper mass 4.88 kg/m² (cold-rolled and lead-coated) or 3.66 kg/m² (shingles, high-yield and lead-coated high-yield)	Direct	Put the specified covering and Table 1404.2 thickness on the wall-type schedule	Verified
Table 1404.2	Masonry and stone coverings	Adhered masonry veneer 6.35 mm ; anchored natural stone 50.4 mm ; architectural cast stone 62.5 mm ; other anchored masonry veneer 66.67 mm ; marble slabs 25.4 mm ; terra cotta anchored 12.7 mm ; terra cotta adhered 9.525 mm	Those coverings specified	Note b exclusive of texture	Conditional	Confirm unit thickness on material submittals against the tabulated minimum	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
Table 1404.2	Metal, glass and panel coverings	Aluminum siding 0.48 mm ; steel (approved corrosion resistant) 0.37 mm ; structural glass 8.73 mm ; glass-fiber reinforced concrete panels 9.525 mm ; porcelain tile 3.125 mm	Those coverings specified	Aluminum siding also AAMA 1402 per 1403.5.1; metal veneer running text uses 0.4 mm sheet steel in 1404.11	Conditional	Coordinate Table 1404.2 thickness with 1404.11 metal-veneer and 1404.12 glass-veneer install rules	Verified
Table 1404.2	Vinyl siding thickness	Vinyl siding 0.88 mm	Vinyl siding specified	None stated in the table row	Conditional	Use only if vinyl is confirmed at podium or low walls; still apply 1404.14 wind limits	Verified
Table 1404.2	Fiber-cement siding thickness	Fiber-cement lap siding 6.35 mm ; fiber-cement panel siding 6.35 mm	Fiber-cement siding specified	Note e is referenced on both rows but the note text is missing from the extract	Conditional	Verify published note e before relying on the fiber-cement thickness rows	Verify source
Table 1404.2	Stucco coat thicknesses	Three-coat and two-coat stucco/exterior cement plaster rows are concatenated; coat thicknesses over metal plaster base, unit masonry and concrete are not recoverable	Stucco or exterior cement plaster specified	Do not invent coat thicknesses	Conditional	Verify the published Table 1404.2 stucco rows before specifying plaster coats; Chapter 25 still governs cement plaster	Verify source

7. Vapor retarders

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.3	Interior vapor retarder	Classify materials per Table 1404.3(1). Provide an interior-side vapor retarder on frame walls per Tables 1404.3(2) and 1404.3(3), or an approved hygrothermal design. Climate zone from IECC Chapter 3 (SBC 601/SBC 602). Class I interior plus Class I exterior requires an approved design	Frame walls	Exception 1: basement walls. Exception 2: below-grade portion of any wall. Exception 3: construction where moisture accumulation/condensation/freezing will not damage materials. Exception 4: Class I and II retarders with vapor permeance greater than 1 perm (ASTM E96 Procedure B) permitted on the interior of any frame wall in all climate zones	Conditional	Apply to framed exterior walls; concrete/masonry backup may fall under Exception 3 only if demonstrated	Verified
1404.3	Class II with exterior foam	Where a Class II interior retarder is combined with foam plastic insulating sheathing as continuous insulation on the exterior, the CI shall comply with Table 1404.3(4) and the Class II retarder shall have vapor permeance greater than 1 perm (ASTM E96 Procedure B)	Class II interior + exterior foam CI on frame walls	Table 1404.3(4) R-values are OCR-destroyed and are not adopted	Conditional	Do not issue CI R-values from Table 1404.3(4) until the published table is checked	Verify source
Table 1404.3(1)	Vapor-retarder classes	Class I: perm rating less than or equal to 0.1 . Class II: greater than 0.1 and less than or equal to 1.0 . Class III: greater than 1.0 and less than or equal to 10	Vapor-retarder classification	Listed deemed-to-comply materials (polyethylene/foil; kraft-faced batts/paint; latex/enamel paint) as tabled	Direct	Identify the interior finish/retarder class on the wall assembly schedule	Verified
Table 1404.3(2)	Interior class by climate zone	Zones 1 and 2 : Class I not permitted, Class II not permitted, Class III permitted. Zone 3 : Class I not permitted, Class II permitted, Class III permitted. Zone 4 except Marine 4: Class I not permitted, Class II permitted, Class III per Table 1404.3(3). Marine 4 and Zones 5–8 : Class I permitted, Class II permitted, Class III per Table 1404.3(3)	Interior vapor retarder on frame walls	Footnote a also points to 1404.3.1 and 1404.3.2	Conditional	Select the row only after the SBC 601/602 climate zone is locked	Verified
Table 1404.3(3)	Class III continuous insulation	Zone 4 : CI R ≥ 0.44 over 50 mm × 100 mm wall or R ≥ 0.66 over 50 mm × 150 mm , or listed vented cladding. Zone 5 : R ≥ 0.88 / R ≥ 1.32 . Zone 6 : R ≥ 0.88 / R ≥ 1.32 . Zone 7 : R ≥ 1.32 / R ≥ 1.32 . R-values are K·m²/W	Class III interior retarder in those zones	Vented cladding includes vinyl lap, polypropylene, horizontal aluminum, brick veneer with airspace, and other approved vented claddings. Insulation for this table does not supersede SBC 601/602 thermal envelope	Conditional	Use vented-cladding or CI branch on framed walls only; do not treat these R-values as the energy-code minimum	Verified
Table 1404.3(3)	Zone 8 Class III CI	Zone 8 CI cells both read R ≥ 2.64 over 50 mm × 100 mm ; the 50 mm × 150 mm pairing is unresolved	Class III in Zone 8	Not a design-release value	Conditional	Verify published Zone 8 cells before any Class III CI in that zone	Verify source
1404.3.1–1404.3.2	Spray-foam Class III moisture control	Spray foam with maximum permeance 1.5 perms at installed thickness on the interior cavity side of WSP, fiberboard, insulating sheathing or gypsum may satisfy Table 1404.3(3) CI where its R-value meets or exceeds the tabulated CI. Combined spray-foam and CI R-values may be counted toward that CI requirement	Class III compliance via cavity spray foam	Does not replace SBC 601/602 thermal checks	Conditional	Record spray-foam perm and R-value on the assembly data sheet if this hybrid path is used	Verified

8. Flashing, weeps and adhered veneer base

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.4	Flashing locations	Install flashing to prevent moisture entering the wall or to redirect it to the finish or to a 1403.2 WRB that is part of 1402.2 drainage. Required at door and window perimeters, penetrations and terminations, wall/roof intersections, chimneys, porches, decks, balconies and similar projections, and built-in gutters. Projecting-flange flashing on both sides and ends of copings, under sills and continuously above projecting trim. Self-adhered fenestration flashings: AAMA 711. Fluid-applied opening flashings: AAMA 714	Exterior wall openings, penetrations and terminations	Tested 1402.2 Exception 2 barrier details may substitute where that path is used	Direct	Detail flashing at every listed condition on wall sections, window/door types and balcony/roof junctions	Verified
1404.4.2	Anchored-veneer weeps	Flashing and weep holes in 1404.6 anchored veneer not more than 250 mm above finished ground level above the foundation wall or slab. At other supports (structural floors, shelf angles, lintels), flashing and weeps in the first course of masonry above the support	Anchored masonry veneer	None stated	Conditional	Dimension foundation weeps and every shelf-angle/lintel flashing course on masonry elevations	Verified
1404.10.1.2.1	Adhered-veneer foundation screed	Corrosion-resistant screed or flashing minimum 0.5 mm or 26 gage galvanized or plastic, minimum vertical attachment flange 90 mm , extending not less than 25 mm below the foundation plate line on exterior stud walls. WRB lapped over the exterior of the attachment flange	Adhered masonry veneer on exterior stud walls	Also comply with 1404.4	Conditional	Show weep-screed gage, flange and extension on the base-of-wall detail	Verified
1404.10.1.3	Adhered-veneer clearances	On exterior stud walls, adhered masonry veneer not less than 100 mm above earth, 50 mm above paved areas, or 12.5 mm above exterior walking surfaces supported by the same foundation as the wall	Adhered masonry veneer on stud walls	None stated	Conditional	Dimension veneer-to-grade and veneer-to-paving clearances on base details	Verified
1404.10.1.4.2–1404.10.1.4.3	Adhered lath and mortar beds	Nominal 12.5 mm scratch coat encapsulating lath, scored horizontally. Nominal 12.5 mm setting bed worked to a nominal 9 mm bed after units are applied. Mortar per 2103 and 2512.2	Adhered masonry veneer installed with lath and mortar	Direct-to-masonry/concrete path per 1404.10.1.5 / 2510.7	Conditional	Specify scratch and setting-bed thicknesses on the adhered-veneer specification	Verified
1404.10.2	Exterior porcelain tile limits	Units weighing more than 0.17 kN/m² : maximum 1200 mm any face dimension, 0.8 m² face area, and not more than 0.30 kN/m² . Units 0.17 kN/m² or less: maximum 1800 mm any face dimension and 1.6 m² face area. Adhere to an approved backing system	Exterior adhered porcelain tile	None stated	Conditional	Check unit weight, edge length and area on the tile schedule before facade use	Verified

9. Combustible cladding on Types I–IV

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1405.1.1	Combustible covering area and height	Combustible exterior wall coverings shall not exceed 10 percent of an exterior wall surface area where fire separation distance is 1500 mm or less, and shall be limited to 12 m in height above grade plane	Types I, II, III and IV construction	Plastics complying with Chapter 26 are excepted from 1405.1. FRTW complying with 2303.2 for exterior installation is not limited in wall area where FSD is 1500 mm or less and is permitted up to 18 m above grade plane regardless of FSD. Wood veneers also comply with 1404.5	Direct	Keep combustible cladding (other than qualifying FRTW) off the tower shaft above 12 m ; map FSD versus combustible percentage on each elevation	Verified
1405.1.1.1	NFPA 268 ignition	Where 1405.1.1 permits combustible exterior wall coverings, test in accordance with NFPA 268	Combustible covering on Types I–IV	Exception 1: wood or wood-based products. Exception 2: other combustible materials covered with an exterior weather covering, other than vinyl sidings, included in and complying with Table 1404.2 thicknesses. Exception 3: aluminum minimum thickness 0.5 mm	Conditional	Require NFPA 268 for combustible finishes that are not wood, Table 1404.2-covered (non-vinyl), or 0.5 mm aluminum	Verified
1405.1.1.1.1	Close-range ignition	Where FSD is 1.5 m or less, combustible exterior wall coverings shall not exhibit sustained flaming as defined in NFPA 268	Combustible covering at FSD 1.5 m or less	None stated beyond 1405.1.1.1 exceptions	Conditional	Record the NFPA 268 no-sustained-flaming result on close FSD elevations	Verified
1405.1.1.1.2	Reduced-flux ignition table	For FSD greater than 1.5 m , covering permitted if exposed to a reduced incident radiant heat flux per NFPA 268 without sustained flaming. Minimum FSD from Table 1405.1.1.1.2 based on the maximum tolerable flux that does not cause sustained flaming	Combustible covering at FSD greater than 1.5 m	Table 1405.1.1.1.2 is not present in the attached extract. Commentary flux/distance examples are not adopted	External verification	Do not adopt a flux-versus-FSD pair until the published table is verified	Verify source

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1405.1.2	Combustible covering at wall top	Combustible exterior wall coverings along the top of exterior walls shall be completely backed up by the exterior wall and shall not extend over or above the top of the exterior wall	Combustible covering at parapet/wall top	None stated	Direct	Stop combustible cladding at the wall top; do not project it as an unbacked cornice above the wall	Verified
1405.1.3	Furred combustible cavity	Where combustible covering is furred out and forms a solid surface, the distance between the back of the covering and the exterior wall shall not exceed 41 mm . The concealed space shall be fireblocked per Section 718	Furred combustible covering forming a solid surface	Distance may exceed 41 mm where the concealed space is not required to be fireblocked by Section 718. Section 718 values are not imported	Conditional	Limit rainscreen cavity behind combustible cladding to 41 mm unless a 718 exception is documented	Verified

10. Selected veneers

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.5	Wood veneer on Types I–IV	Not less than 25 mm nominal thickness, 10 mm exterior hardwood siding, or 9 mm exterior-type wood structural panels or particleboard. Height not more than 12 m above grade (18 m where fire-retardant-treated wood is used). Attached to or furred from noncombustible backing with the required fire-resistance rating. Open or spaced wood veneers without concealed spaces shall not project more than 600 mm	Wood veneer on Type I, II, III or IV walls	None stated beyond the FRTW height increase	Conditional	Confine wood veneer to the stated height band and show projection and noncombustible backing on details	Verified
1404.6–1404.6.2	Anchored masonry veneer	Comply with 1404.6 through 1404.9 and SBC 305 Sections 12.1 and 12.2. Chapter 14 anchored veneers are not required to meet SBC 305 Article 3.3 F1 tolerances. Seismic Design Category C, D, E or F: SBC 305 Section 12.2.2.11	Anchored masonry veneer	SBC 305 numeric detailing is not imported	Conditional	Note SBC 305 as the veneer design standard; keep Chapter 14 flashing/weep and thickness checks on the drawings	Verified
1404.7 Item 1	Stone veneer on masonry/concrete	Units not exceeding 250 mm thick. Ties not less than 2.7 mm wire; loop legs not less than 150 mm ; eyes 300 mm maximum on center both ways; 2.7 mm wire tie for every 0.2 m² ; tie legs not less than 375 mm with last 50 mm bent; 25 mm minimum cement grout between backing and veneer	Anchored stone veneer on concrete or masonry backing	None stated	Conditional	Put tie size, spacing, grout thickness and unit cap thickness on the stone-veneer typical	Verified
1404.7 Items 2–3	Stone veneer on studs	50 by 50 mm , 1.6 mm mesh; two WRB layers; studs not more than 400 mm on center; mesh fasteners 100 mm on center and 200 mm into plates/tracks; wood: 50 mm nails with 30 mm penetration; steel: #8 screws with not fewer than three exposed threads; 2.7 mm tie every 0.2 m² with 375 mm legs and 50 mm bends; 25 mm grout; steel studs minimum bare thickness 1 mm	Anchored stone veneer on wood or cold-formed steel studs	Page split after “minimum 30” continues as 30 mm penetration	Conditional	Use the stud-backing typical only if stone is on framed walls; show two WRB layers and grout thickness	Verified
1404.8	Slab-type stone veneer	Units not exceeding 50 mm thick; dowels in the middle third of edges, not more than 600 mm apart around the periphery, not fewer than four ties per unit; unit area not more than 2 m² . Loose dowel holes not more than 1.6 mm oversize, countersink diameter and depth equal to twice the dowel diameter. Ties resist two times the veneer weight; sheet-metal ties not smaller than 0.853 by 25 mm ; wire not smaller than 3.70 mm	Anchored slab marble, travertine, granite or similar	None stated	Conditional	Limit slab size and show four-edge dowel spacing on each unit type	Verified
1404.9	Terra cotta veneer	Units not less than 40 mm thick; dovetail webs approximately 200 mm on center. Anchors not less than No. 8 gage at the top of each piece in bed joints not less than 300 mm nor more than 450 mm on center, secured to 6 mm pencil rods. Facing set not less than 50 mm from backing, solid-filled with Portland cement grout and pea gravel. Ties support the full weight of the veneer in tension	Anchored terra cotta or ceramic units	None stated	Conditional	Show 50 mm grouted cavity, web spacing and bed-joint anchor spacing on the terra-cotta typical	Verified
1404.11–1404.11.1	Metal veneer	Sheet steel not less than 0.4 mm nominal thickness. Fastenings or ties not more than 600 mm vertically or horizontally; units exceeding 0.4 m² have not fewer than four attachments per unit. Attachments not less than W 1.7 wire. Design attachments for Chapter 16 / SBC 301 wind (outbound)	Exterior metal veneer	Corrosion-resistant or porcelain-enamel protection required	Conditional	Dimension fastener grid and panel area on metal-panel shop drawings	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.12–1404.12.2	Glass veneer size	Single section not more than 1.0 m² where not more than 4500 mm above sidewalk/grade, and not more than 0.56 m² where more than 4500 mm above that level. Length or height not more than 1200 mm . Thickness not less than 8 mm	Thin exterior structural glass veneer	None stated	Conditional	Check each glass-veneer unit area, edge and thickness against height above grade	Verified
1404.12.3–1404.12.6	Glass veneer support	At least 50 percent of each unit bonded by mastic 6 mm to 16 mm thick. At sidewalk: metal molding, glass 6 mm above highest sidewalk point. Above bulkhead or more than 900 mm above sidewalk: nonferrous shelf angles not less than 1 mm thick and 50 mm long, not fewer than two per unit. Horizontal joints not less than 1.6 mm ; expansion joints not less than 6 mm where abutting nonresilient material. Above show-window heads or more than 3600 mm above sidewalk: mechanical fastenings at each edge or four corners in addition to mastic and shelf angles	Thin exterior structural glass veneer	Building official may approve other jointing	Conditional	Combine mastic thickness, shelf angles and mechanical fixings on the glass-veneer section by height zone	Verified

11. Light cladding at podium

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1404.14	Vinyl siding wind pressure	Vinyl siding ASTM D3679 permitted where design wind pressure per 1609 does not exceed 1.45 kN/m² . Where pressure exceeds 1.45 kN/m² , tests or calculations indicating Chapter 16 compliance shall be submitted	Vinyl siding specified	Manufacturer instructions may control fasteners	Conditional	Do not use the 1.45 kN/m² deemed-to-comply path on a high-rise wind exposure without Chapter 16 evidence	Verified
1404.14.1.1–1404.14.1.3	Vinyl siding fasteners	Corrosion-resistant nails, head not less than 7.9 mm , shank not less than 3.18 mm , penetration not less than 32 mm into nailable substrate. Cold-formed steel: not fewer than three exposed threads. Spacing not greater than 400 mm for horizontal siding and 300 mm for vertical siding, fasteners in the middle third of nail-hem slots	Vinyl siding on wood or CFS	Approved manufacturer instructions may specify otherwise	Conditional	Put fastener size, penetration and spacing on the vinyl typical if used at podium	Verified
1404.16–1404.16.2	Fiber-cement siding	ASTM C1186 Type A minimum Grade II (or ISO 8336 Category A minimum Class 2). Wood stud nails penetrate not less than 25 mm . CFS screws not fewer than three exposed full threads. Lap siding maximum width 300 mm , lapped minimum 32 mm	Fiber-cement panel or lap siding	Manufacturer instructions may specify otherwise	Conditional	Show lap, joint protection and fastener penetration on fiber-cement details	Verified
1403.12.2	Polypropylene siding FSD	Fire separation distance between a building with polypropylene siding and the adjacent building not less than 3 m	Polypropylene siding used	Flame-spread specimen must remain in position during ASTM E84 or UL 723 per 1403.12.1	Conditional	Do not place PP siding on walls with FSD under 3 m	Verified
1404.18	Polypropylene siding wind/height	Limited to areas where Chapter 16 wind speed does not exceed 45 m/s and building height is less than or equal to 12 m in Exposure C. Where wind exceeds 45 m/s or height exceeds 12 m , tests or calculations indicating Chapter 16 compliance shall be submitted	Polypropylene siding specified	Manufacturer installation instructions still apply	Conditional	Treat PP siding as a low-height product; the tower shaft exceeds the 12 m deemed-to-comply height	Verified

12. Metal composite materials (MCM)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1406.9	General MCM surface burning	Unless otherwise specified, MCM flame spread index 75 or less and smoke-developed index 450 or less, ASTM E84 or UL 723, maximum thickness intended for use	MCM used	Types I–IV use the stricter 1406.10.1 indices	Conditional	Record E84/UL 723 indices at the installed thickness on the MCM specification	Verified
1406.10	MCM height split	Types I–IV: comply with 1406.10.1 and 1406.10.2 for installations up to 12 m above grade plane. Comply with 1406.10.1 through 1406.10.3 for installations greater than 12 m above grade plane	MCM on Type I, II, III or IV	Type V permission in 1406.11 is omitted here as not typical for the tower shaft	Direct	Treat tower MCM as the greater-than- 12 m path, including NFPA 285	Verified
1406.10.1	Types I–IV MCM surface burning	Flame spread index not more than 25 and smoke-developed index not more than 450 , maximum thickness intended for use, ASTM E84 or UL 723	MCM on Types I–IV	None stated	Direct	Specify FSI 25 / SDI 450 for tower MCM, not the general 75 / 450 pair	Verified

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1406.10.2	MCM thermal barrier	Separate MCM from the interior by 12.5 mm gypsum wallboard or a material meeting both NFPA 275 Temperature Transmission and Integrity Fire Tests	MCM on Types I–IV	Exception 1: MCM system specifically approved from NFPA 286 with 803.1.1.1 criteria, UL 1040 or UL 1715, maximum thickness, with typical seams/joints. Exception 2: MCM as balcony/projection elements, architectural trim or embellishments	Direct	Show the 12.5 mm thermal barrier or a listed exception on the MCM wall section	Verified
1406.10.3	MCM NFPA 285	MCM system tested to and complying with NFPA 285, MCM in the maximum thickness intended for use	Types I–IV MCM greater than 12 m above grade plane	None stated in this section	Direct	Require an NFPA 285 listing for the full MCM wall assembly at maximum thickness	Verified
1406.8	MCM on rated walls	Where MCM is used on exterior walls required to have a fire-resistance rating per Section 705, submit evidence that the required rating is maintained	MCM on a fire-resistance-rated exterior wall	Exception: MCM without foam plastic, on the outer surface, attachments not penetrating through the entire exterior wall assembly	Conditional	Coordinate rated-wall listings with Chapter 7; do not import Table 705 values here	Verified
1406.12	MCM with foam	Where MCM systems are in an exterior wall envelope containing foam plastic insulation, the envelope shall also comply with Section 2603	Foam and MCM in the same envelope	Chapter 26 values are not imported	Conditional	Flag any foam + MCM wall as a Chapter 26 assembly, not MCM-only	Verified

13. Exterior insulation and finish systems (EIFS)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1407.2–1407.4	EIFS performance	Construct EIFS to ASTM E2568. Framing/substrate resist Chapter 16 loads. EIFS complies with 1402 weather resistance and manufacturer instructions	EIFS specified	Special inspections per 1704.2 and 1705.17	Conditional	List ASTM E2568 and Chapter 17 special inspections on the EIFS specification	Verified
1407.4.1	EIFS with drainage	Average minimum drainage efficiency 90 percent per ASTM E2273. Required on framed walls of Type V construction, Group R-1, R-2, R-3 and R-4 occupancies	Type V framed walls in Group R, or EIFS with drainage otherwise specified	WRB for drainage EIFS: 1403.2 or ASTM E2570	Conditional	If any Type V framed R-2 wall exists (podium/townhouse-style), specify drainage EIFS at 90 percent ; do not assume Type V for the tower shaft	Verified

14. High-pressure laminates (HPL)

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1408.9	General HPL surface burning	Unless otherwise specified, HPL flame spread index 75 or less and smoke-developed index 450 or less, ASTM E84 or UL 723, minimum and maximum thicknesses intended for use	HPL used	Types I–IV use 1408.10 or the 1408.11 alternate	Conditional	Record min-and-max thickness burns on the HPL specification	Verified
1408.10–1408.10.1	Types I–IV HPL surface burning	Comply with 1408.10.1 through 1408.10.4, or with 1408.11. HPL flame spread index not more than 25 and smoke-developed index not more than 450 , min and max thicknesses, ASTM E84 or UL 723	HPL on Type I, II, III or IV	1408.11 alternate is a separate height/area path	Direct	Default the tower to 1408.10 (including NFPA 285), not the 12 m alternate	Verified
1408.10.2–1408.10.3	HPL thermal barrier	Separate HPL from the interior by 12.5 mm gypsum wallboard or NFPA 275 (both tests). Thermal barrier not required where the HPL system is approved from NFPA 286 with 803.1.1.1, UL 1040 or UL 1715 (min and max thicknesses, typical details), or where HPL is balcony/projection, trim or embellishment	HPL on Types I–IV	Listed large-scale tests or trim/balcony use	Direct	Show the 12.5 mm barrier or a documented 1408.10.3 exception	Verified
1408.10.4	HPL NFPA 285	HPL system tested to and complying with NFPA 285, HPL in the minimum and maximum thicknesses intended for use	Types I–IV HPL on the 1408.10 path	1408.11 alternate may omit 1408.10.1–1408.10.4 if its limits are met	Direct	Require NFPA 285 at both min and max HPL thickness for tower installations	Verified
1408.11–1408.11.1.2	HPL 12 m alternate	HPL not required to comply with 1408.10.1–1408.10.4 where installed up to 12 m above grade plane and: FSD 1.5 m or less, HPL area not more than 10 percent of the exterior wall surface; or FSD greater than 1.5 m , no HPL area limit	Alternate to full 1408.10 on Types I–IV	Height still capped at 12 m on this path. Type V permission in 1408.12 omitted as not typical for the tower shaft	Conditional	Use the alternate only for HPL that stays at or below 12 m ; tower elevations above that remain on 1408.10	Verified
1408.8	HPL on rated walls	Where HPL is used on exterior walls required to have a fire-resistance rating per Section 705, submit evidence that the required rating is maintained	HPL on a fire-resistance-rated exterior wall	Exception: HPL without foam plastic, outer surface, attachments not through the entire wall	Conditional	Coordinate rated-wall evidence with Chapter 7	Verified
1408.13	HPL with foam	HPL systems containing foam plastic insulation shall also comply with Section 2603	Foam in the HPL system/envelope	Chapter 26 values are not imported	Conditional	Flag foam-cored HPL as a Chapter 26 assembly	Verified

15. Outbound controls

Cite	Component / check	Exact source requirement / value	Trigger / condition	Exceptions / branch	R-2 tower status	Design action	Source confidence
1402.3	Structural loads	Exterior walls and associated openings shall resist superimposed loads required by Chapter 16	All exterior walls and openings	No Chapter 16 values imported	External verification	Size cladding, anchors and openings from Chapter 16 / SBC 301, not from this matrix	Verified
1402.4	Fire resistance and openings	Exterior walls fire-resistance rated as required by other sections, with opening protection as required by Chapter 7	Exterior walls	Commentary names Tables 601, 602 and 705.8; those values are not in this chapter's code text and are not adopted	External verification	Pull ratings and unprotected-opening percentages from Chapters 6 and 7	Verified
1402.6–1402.7	Flood resistance	In flood hazard areas, exterior walls below the 1612 elevation use flood-damage-resistant materials. In coastal high-hazard areas and coastal A zones, electrical, mechanical and plumbing components shall not be mounted on or penetrate breakaway exterior walls	Flood hazard / coastal A / coastal high-hazard designation	Section 1612 elevations and ASCE 24 are not imported	Conditional	Confirm 1612 flood designation before detailing below-elevation walls	Verified
1403.3–1403.8	Material chapters	Wood walls: Chapter 23. Masonry: Chapter 21. Steel: Chapter 22. Aluminum: Chapter 20. Concrete: Chapter 19. Glass-unit masonry: Chapter 21. Plastic panels: Chapter 26 plus Chapter 16 wind	Those materials used	No outbound chapter values imported	External verification	Keep material design in the named chapters; use Chapter 14 for covering, WRB and combustible-cladding checks	Verified
1403.13–1403.14	Foam plastic and through-foam attachment	Foam plastic insulation in exterior wall covering assemblies shall comply with Chapter 26. Coverings attached through foam plastic insulating sheathing shall comply with 2603.11, 2603.12 or 2603.13	Foam in the wall or cladding fastened through foam sheathing	Chapter 26 numeric attachment/NFPA 285 rules are not imported	External verification	Route foam walls and through-foam fasteners to Chapter 26	Verified
1404.13–1404.13.1	Exterior windows and doors	Windows and doors in exterior walls conform to testing and performance in Section 1705.9. Install per approved manufacturer instructions; fastener size and spacing from those instructions based on maximum loads and spacing used in the tests	Exterior fenestration	No 1705.9 pressures imported	External verification	Put 1705.9 test reports on the fenestration schedule; keep 1404.4 flashing in the wall details	Verified
1404.15	Cement plaster	Exterior cement plaster conforms to Chapter 25	Stucco / cement plaster	Gypsum plaster is not an exterior covering in this clause	External verification	Detail stucco coats from Chapter 25 after verifying Table 1404.2 stucco rows	Verified
1404.17	Weatherboard fastening	Secure weather boarding and wall coverings with approved corrosion-resistant fasteners per Table 2304.10.2 or approved manufacturer instructions, except where wood sheathing is not less than 25 mm nominal thickness	Weather boarding / wall coverings	Table 2304.10.2 nail schedules are not imported	External verification	Use Chapter 23 nailing tables for wood coverings	Verified
1409.1	Plastic composite decking	Exterior deck boards, stair treads, handrails and guards of plastic composites, including plastic lumber, shall comply with Section 2612	Plastic composite decking, treads, handrails or guards	Section 2612 values are not imported	Conditional	Route balcony deck boards and composite guards to 2612	Verified

16. Project-use controls

1. Use **Verified** rows for initial facade coordination after the row trigger, construction type and cladding system are confirmed.
2. Treat every **Verify source** row as a design hold point. No affected value is to be placed in issued-for-approval drawings without a published-source check.
3. Do not import Chapter 7 opening percentages, Chapter 16 wind pressures, Chapter 26 foam/NFPA 285 acceptance criteria, SBC 305 veneer tables, or SBC 601/602 climate-zone assignments.
4. Do not treat the occupied-floor height above **23 m** as permission to run combustible cladding, vinyl, or polypropylene up the tower shaft. Types I–IV combustible covering remains capped at **12 m (18 m FRTW)** unless a different Chapter 14 path (MCM/HPL NFPA 285) is documented.
5. Do not use Table 1404.3(4) R-values or Table 1405.1.1.1.2 flux/FSD pairs from commentary or memory.
6. Record construction type, cladding, WRB combustibility, foam, climate zone and FSD in the project Golden Thread. This matrix is not evidence of SCD NOC or stamped compliance.

17. Coverage summary

Internal inventory of the attached Chapter 14 extract (numbered code, exceptions, tables, footnotes; commentary excluded). Row-level records are not published.

- **Inventory scope:** numbered code, exceptions, tables, footnotes (commentary excluded)
- **Total independently checkable numeric records:** 225
- **Verified:** 213
- **Verify source:** 12
- **Numeric records in Section 1401:** 0
- **Numeric records in Section 1409:** 0 (outbound to 2612 only)

Counts by top-level section

Top-level section	Records
1401	0
1402	15
1403	3
1404	180
1405	10
1406	7
1407	1
1408	9
1409	0

Appended-table coverage

Appended table	Records	Verify source records
Table 1404.2	36	8
Table 1404.3(1)	5	0
Table 1404.3(2)	0	0
Table 1404.3(3)	10	2
Table 1404.3(4)	1	1
Table 1405.1.1.1.2	1	1

Coverage cross-check against SBC 201 Chapter 14 Exterior Walls (2024)_CS.md was topics-only: weather envelope and WRB; vapor retarders; combustible cladding on Types I–IV; MCM/HPL/EIFS/foam. No CS.md value was copied into a matrix cell. Figure 1404.3(1) climate maps, Figure 1404.3(2) IECC thermal definitions, and other commentary figures were not inventoried.

18. Unresolved-source register

Hold points for the 12 **Verify source** inventory records. Counts are record counts, not distinct numeric values. No value in this register is a design-release figure.

Affected table / clause	Why unverified	Control note
Table 1404.2 stucco rows	5 three-coat/two-coat thickness cells concatenated; coat thicknesses over metal plaster base, unit masonry and concrete not recoverable	Verify published plaster-coat thicknesses; do not invent coats. Chapter 25 still governs cement plaster
Table 1404.2 note e	3 products (fiber-cement lap, fiber-cement panel, hardboard) cite note e; note e text is absent from the extract	Verify published note e before using those thickness rows as complete
1404.3 / Table 1404.3(4)	Entire continuous-insulation + Class II table is OCR-destroyed (R - va l u e ≥ loop); no cell adopted	Do not reconstruct CI R-values. Lock climate zone from SBC 601/602, then read the published table
Table 1404.3(3) Zone 8	2 CI cells both print R ≥ 2.64 over 50 mm × 100 mm; 50 mm × 150 mm pairing unresolved	Verify published Zone 8 row before Class III CI in that zone
Table 1405.1.1.1.2	Table cited by 1405.1.1.1.2 is not in the extract; commentary 9.5 kW/m² → 30 m example is not adopted	Verify published flux-versus-FSD table before any reduced-flux NFPA 268 placement